StellarNet

CDemo SDK: C++

Documentation v1.0



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INTRODUCTION

CDemoSDK demo tutorial for Microsoft Visual Studio.

Demo Folder Contains:

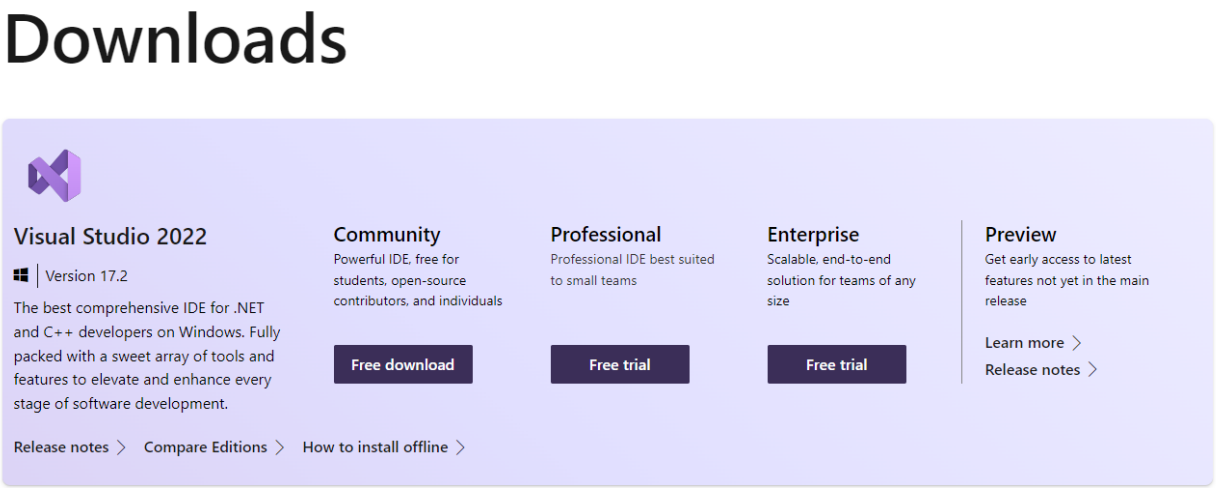
- CDemo.vcxproj & CDemo.vcproj
  - VC++ project files
- Source Codes
  - CDemo.cpp
    - Demo file of using StellarNet spectrometer functions.
  - StellarNet.cpp
    - Interface to swdll.dll/swdll64.dll
- Header files
  - CDemo.h
  - StellarNet.h
- Swdll64.dll & swdll.dll
  - DLL files
- README.md
  - A quick instruction on how to run the provided demo program.
- Other Microsoft Visual Studio project related files.

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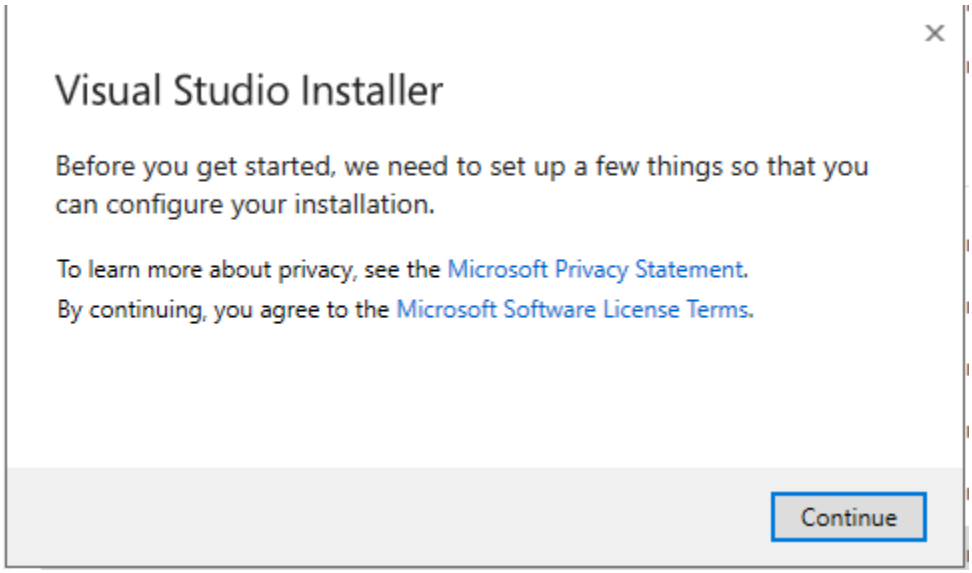
MICROSOFT VISUAL STUDIO INSTALLATION

Visit the link below to down Microsoft Visual Studio.

https://visualstudio.microsoft.com/downloads/

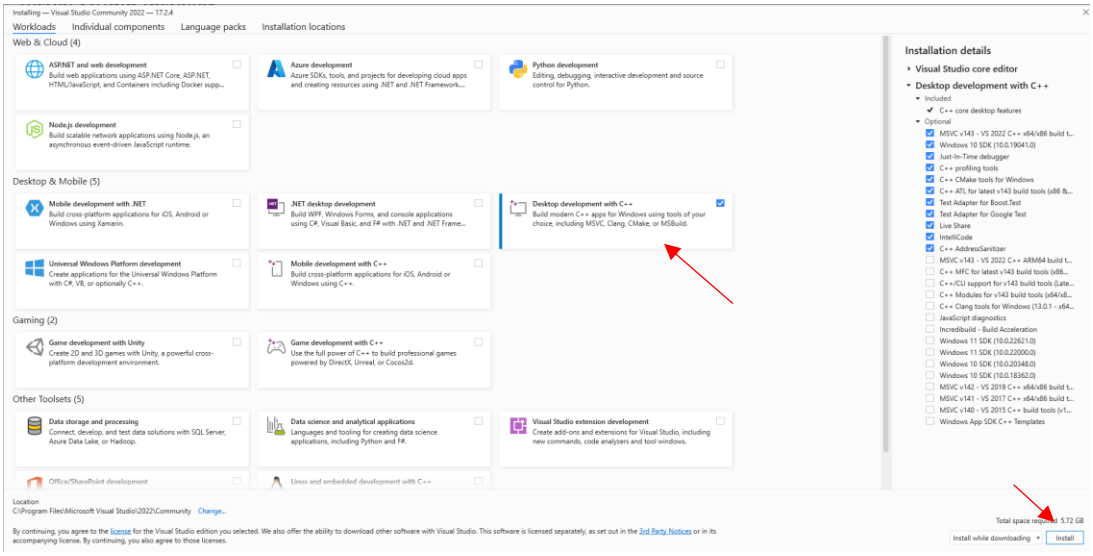


- Double click on the VisualStudioSetup to install the application.
- Click “Yes” to allow the app to make change on your device.
- Click “Continue” to start the installation.



- Select “Desktop development with C++” and click “Install”

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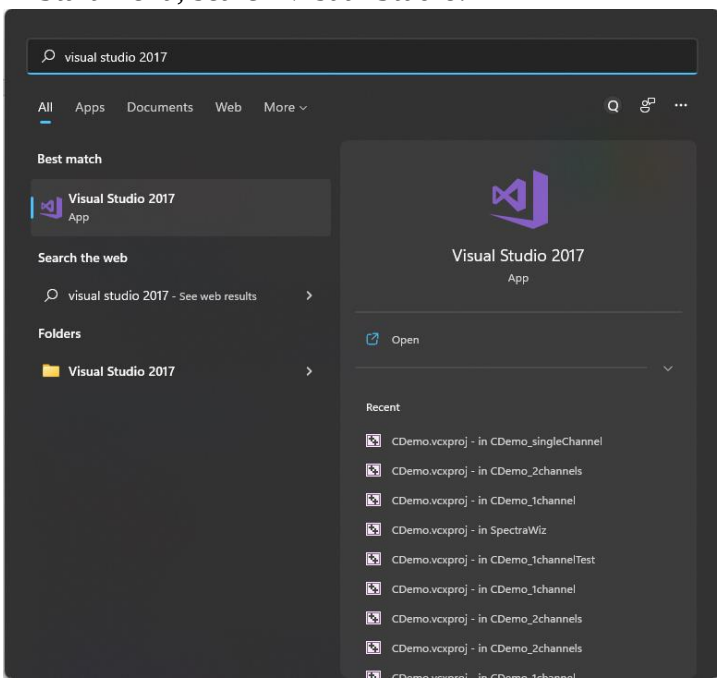


Note, if you get an import error or if you have any questions, you can email Support@StellarNet.us with screenshots to illustrate the situation.

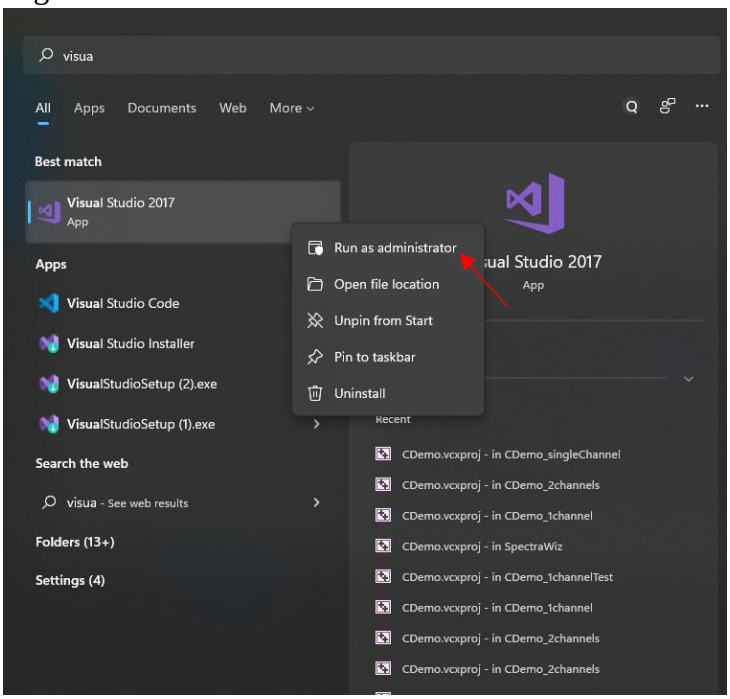
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RUN DEMO PROGRAM

- Demo program must be run with the Microsoft Visual Studio as an administrator.
  - In Start menu, search Visual Studio.

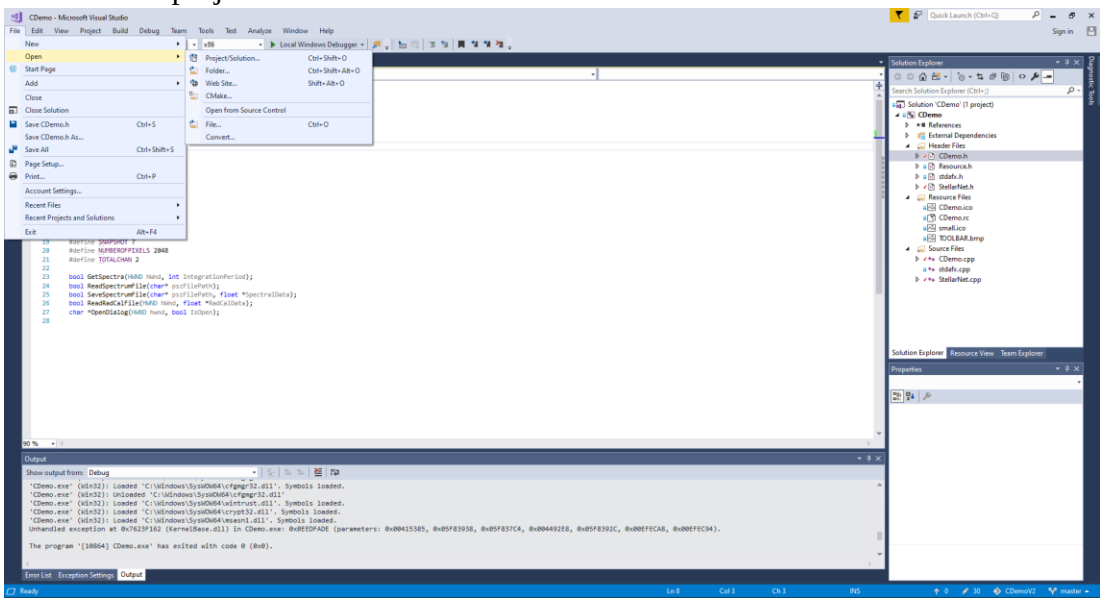


  - Right click on the Visual Studio and select “Run as administrator”.

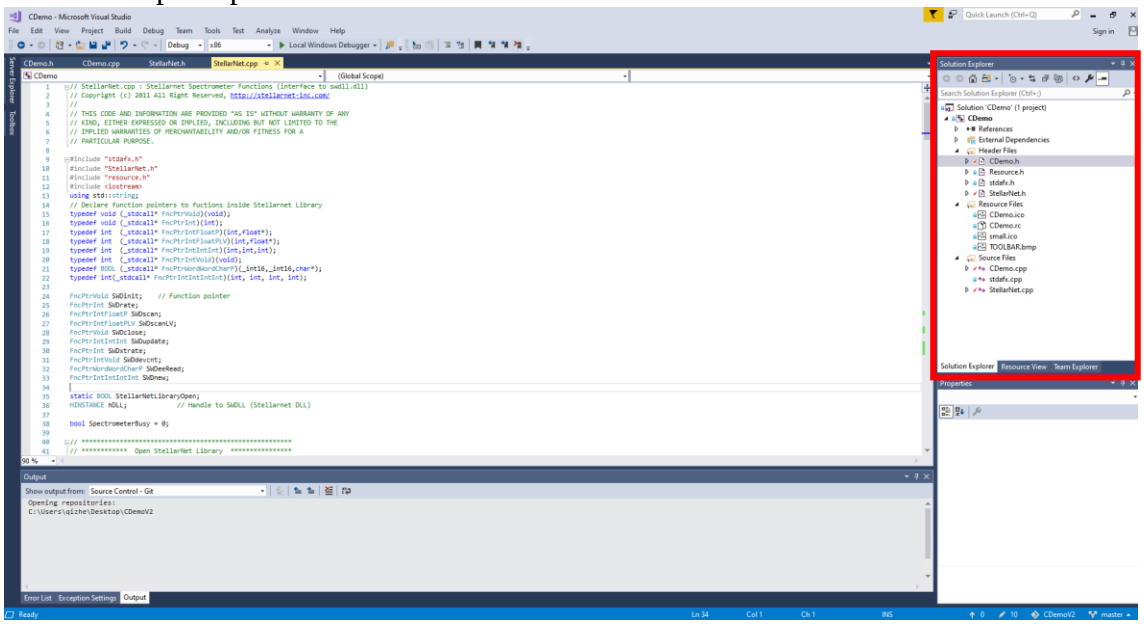


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- Click “File” → “Open” → “Project/Solution…” to open project file “CDemo.vcxproj”.

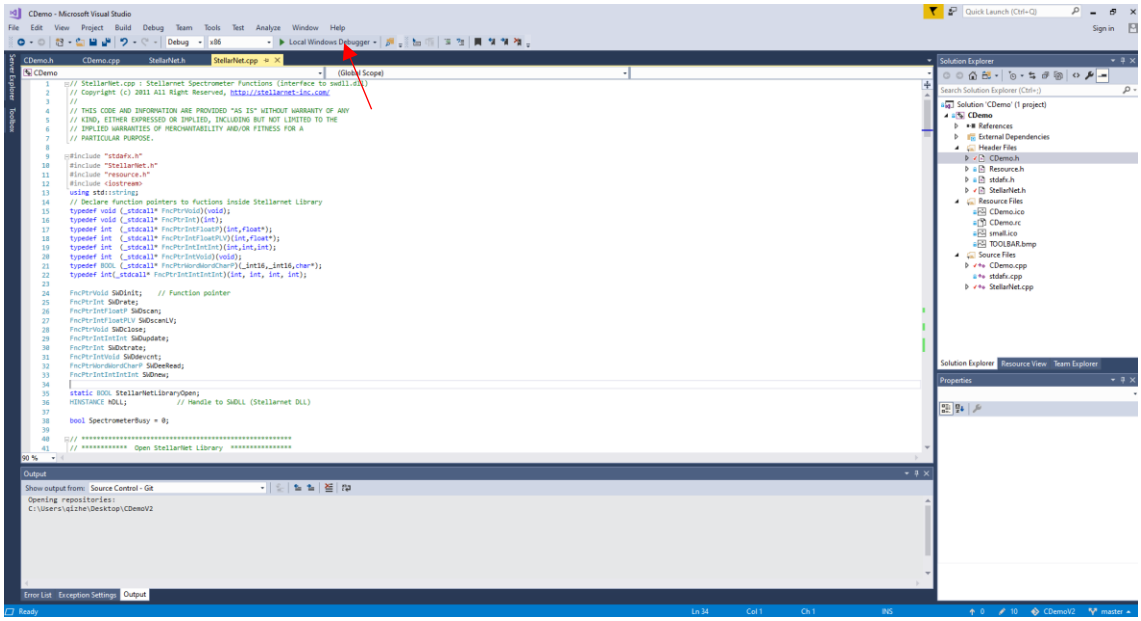


- You can find all the source code imported into the Visual Studio Project in the Solution Explore panel.



- Plug in the spectrometer through USB and then click “Local Windows Debugger” to run the demo program.

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SWDLL.DLL LIBRARY

SWDini()

Startup (call only once).

External 'swdll' Index: 1

Parameters: NONE

Return: NONE

Example:

```
typedef void (_stdcall* FncPtrVoid)(void);

FncPtrVoid SWDinit;    // Function pointer

hDLL = LoadLibrary("swdll64.dll");

// Put adresses of SWDLL functions into function pointers.
SWDinit = (FncPtrVoid)GetProcAddress(hDLL, "SWDinit");

SWDinit();    //Call only once!!!!
```

SWDscan(chan, buffers)

Acquire spectral scan from channels 1 to 8.

External 'swdll' Index: 2

Parameters:

Name	Data Type	Required/Optional	Description
chan	Integer	Required	Channel to perform the scanning
buffers	aptype	Required	Data buffer

Return: Integer

Name	Description
0	Success
1	Busy
2	Timeout
3	Fatal (nil buffer)

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Example:

```
typedef int  (_stdcall* FncPtrIntFloatP)(int,float*);

FncPtrIntFloatP SWDscan     // Function pointer

hDLL = LoadLibrary("swdll64.dll");

// Put adresses of SWDLL functions into function pointers.
SWDscan = (FncPtrIntFloatP)GetProcAddress(hDLL, "SWDscan");

scode = SWDscanLV( chan, DataBuffer);  // get spectral scan
```

SWDrate(dsf)

Change detector integration rate in milli-seconds (4-65500).

External ‘swdll’ Index: 3

Parameters:

Name	Data Type	Required/Optional	Description
dsf	Integer	Required	Detector integration rate in milli-seconds. Valid range is from 4 to 65500.

Return: NONE

Example:

```
typedef void (_stdcall* FncPtrInt)(int);

FncPtrInt SWDrate;

hDLL = LoadLibrary("swdll64.dll");

// Put adresses of SWDLL functions into function pointers.
SWDrate = (FncPtrInt)GetProcAddress(hDLL, "SWDrate");

SWDrate(IntegrationPeriod);
```

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SWDclose()

Shutdown (call only once).

External 'swdll' Index: 4

Parameters: NONE

Return: NONE

Example:

```
typedef void (_stdcall* FncPtrVoid)(void);

FncPtrVoid SWDclose;

hDLL = LoadLibrary("swdll64.dll");

// Put adresses of SWDLL functions into function pointers.
SWDclose = (FncPtrVoid)GetProcAddress(hDLL, "SWDclose");

SWDclose();             // allow SWDLL to release its resources
```

SWDupdate(ScansToAverage, xsmooth, TempComp)

Dynamic update for data processing configuration.

External 'swdll' Index: 5

Parameters:

Name	Data Type	Required/Optional	Description
ScansToAvg	Integer	Required	Scan averaging accumulates scans then divides by ScansToAvg.
Xsmooth	Integer	Required	Smoothing uses pixel boxcar averaging across spectral scan. 0 = none 1 = 5_pixel 2 = 9_pixel 3 = 17_pixel 4 = 33_pixel
TempComp	Boolean	Required	Temperature compensation uses detector optical black region pixels 1-12. 0 = none 1 = on

Return: NONE

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Example:

```
typedef int    (_stdcall* FncPtrIntIntInt)(int,int,int);

FncPtrIntIntInt SWDupdate;

hDLL = LoadLibrary("swdll64.dll");

// Put adresses of SWDLL functions into function pointers.
SWDupdate = (FncPtrIntIntInt)GetProcAddress(hDLL, "SWDupdate");

SWDupdate(ScansToAvg, xsmooth, TemperatureCompensation);
```

SWDxtmode(xtmode)

Set xtiming OEM mode for capable units 0 = off and 1 = on (RESERVED for OEMs).

External 'swdll' Index: 6

Parameters:

Name	Data Type	Required/Optional	Description
xtmode	Integer	Required	Xtiming mode. 0 = off 1 = on

Return: NONE

Example:

```
typedef int    (_stdcall* FncPtrInt)(int);

FncPtrInt SWDxtmode;

hDLL = LoadLibrary("swdll64.dll");

// Put adresses of SWDLL functions into function pointers.
SWDxtmode = (FncPtrInt)GetProcAddress(hDLL, "SWDxtmode");

SWDxtmode(xtmode);
```

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SWDxtrate(xtrate)

Set xtiming resolution clock rate 0-2 where 0 = fastest and lowest resolution.

External 'swdll' Index: 7

Parameters:

Name	Data Type	Required/Optional	Description
xtrate	Integer	Required	Xtiming resolution clock rate. Range from 0 to 2, where 0 is the fastest and lowest resolution.

Return: NONE

Example:

```
typedef int    (_stdcall* FncPtrInt)(int);

FncPtrInt SWDxtrate;

hDLL = LoadLibrary("swdll64.dll");

// Put adresses of SWDLL functions into function pointers.
SWDxtrate = (FncPtrInt)GetProcAddress(hDLL, "SWDxtrate");

SWDxtrate(xtrate);
```

SWDscanLV(chan, buffers)

Acquire spectral similar to SWDscan for LabView and C++ programs. Caller allocates buffer then call to SWDscanLV fills it.

External 'swdll' Index: 8

Parameters:

Name	Data Type	Required/Optional	Description
chan	Integer	Required	Channel to perform the scanning.
buffers	aptype	Required	Data buffer

Return: Integer

Name	Description
0	Success
1	Busy
2	Timeout
3	Fatal (nil buffer)

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Example:

```
typedef int    (_stdcall* FncPtrInt)(int);

FncPtrIntFloatPLV SWDscanLV;

hDLL = LoadLibrary("swdll64.dll");

// Put adresses of SWDLL functions into function pointers.
SWDscanLV = (FncPtrIntFloatPLV)GetProcAddress(hDLL, "SWDscanLV");

SWDscanLV( chan, DataBuffer)
```

SWDdevcnt()

Return the number of spectrometers currently attached.

External 'swdll' Index: 9

Parameters: NONE

Return: Number of spectrometers attached (Integer).

Example:

```
typedef int    (_stdcall* FncPtrIntVoid)(void);

FncPtrIntVoid SWDdevcnt;

hDLL = LoadLibrary("swdll64.dll");

// Put adresses of SWDLL functions into function pointers.
SWDdevcnt = (FncPtrIntVoid)GetProcAddress(hDLL, "SWDdevcnt");

int NumberOfSpectrometers = SWDdevcnt();
```

SWDeeRead(chan, eaddr, buffer)

Read data from spectrometer EEROM at a specific address.

External 'swdll' Index: 10

Parameters:

Name	Data Type	Required/Optional	Description
chan	Integer	Required	Spectrometer channel.
eaddr	_int16	Required	EEROM address.
buffers	etype	Required	Data buffer

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Return: EEROM read is successful or not (Boolean). True for successful and False for unsucessful.

Example:

```
typedef int    (_stdcall* FncPtrWordWordCharP)(void);

FncPtrIntVoid SWDeeRead;

hDLL = LoadLibrary("swdll64.dll");

// Put adresses of SWDLL functions into function pointers.
SWDeeRead = (FncPtrWordWordCharP)GetProcAddress(hDLL, "SWDeeRead");

SWDeeRead(ec, ea, buffer)
```

SWDeeWrite(chan, eaddr, buffer)

Write data to spectrometer EEROM at a specific address.

External 'swdll' Index: 11

Parameters:

Name	Data Type	Required/Optional	Description
chan	Integer	Required	Spectrometer channel.
eaddr	_int16	Required	EEROM address in hexadecimal.
buffers	etype	Required	Data buffer

Return: EEROM read is successful or not (Boolean). True for successful and False for unsucessful.

Example:

```
typedef int    (_stdcall* FncPtrWordWordCharP)(void);

FncPtrIntVoid SWDeeWrite;

hDLL = LoadLibrary("swdll64.dll");

// Put adresses of SWDLL functions into function pointers.
SWDeeWrite = (FncPtrWordWordCharP)GetProcAddress(hDLL, "SWDeeWrite");

SWDeeWrite(ec, ea, buffer)
```

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SWDloadMap()

Load hotpix map for all 8 channels.

External ‘swdll’ Index: 12

Parameters: NONE

Return: NONE

Example:

```
typedef void (_stdcall* FncPtrVoid)(void);

FncPtrVoid SWDloadMap;    // Function pointer

hDLL = LoadLibrary("swdll64.dll");

// Put adresses of SWDLL functions into function pointers.
SWDloadMap = (FncPtrVoid)GetProcAddress(hDLL, "SWDloadMap");

SWDloadMap ();       //Call only once!!!!
```

SWDfifo()

Start big FIFO. Count down from 31 to 0 or 127 to 0 if is 256k.

External ‘swdll’ Index: 13

Parameters: NONE

Return: NONE

Example:

```
typedef void (_stdcall* FncPtrVoid)(void);

FncPtrVoid SWDfifo;    // Function pointer

hDLL = LoadLibrary("swdll64.dll");

// Put adresses of SWDLL functions into function pointers.
SWDfifo = (FncPtrVoid)GetProcAddress(hDLL, "SWDfifo");

SWDfifo ();   //Call only once!!!!
```

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SWDnew(pfunc, pval1, pval2, pval3, pval4)

Set DigPots and big FIFO.

External 'swdll' Index: 14

Parameters:

Name	Data Type	Required/Optional	Description
pfunc	Integer	Required	0 = Gain pot 1 = Bias pot 2 = fiflgs 3 = ficnt 31/127
pval1	Integer	Required	0 to 255 plot / FtopMSB assume LSB = 0.
pval2	Integer	Required	FtopLSB
pval3	Integer	Required	Spare

Return: NONE

Example:

```
typedef int(_stdcall* FncPtrIntIntIntInt)(int, int, int, int);

FncPtrIntIntIntInt SWDnew;

hDLL = LoadLibrary("swdll64.dll");

// Put adresses of SWDLL functions into function pointers.
SWDnew = (FncPtrIntIntIntInt)GetProcAddress(hDLL, "SWDnew");

SWDnew(pFunc, pval1, pval2, pval3);
```